Geomorphic and Soil Controls on Berm Structural Integrity and Vegetation Response in the Altar Valley, Arizona

Figure Registry Report

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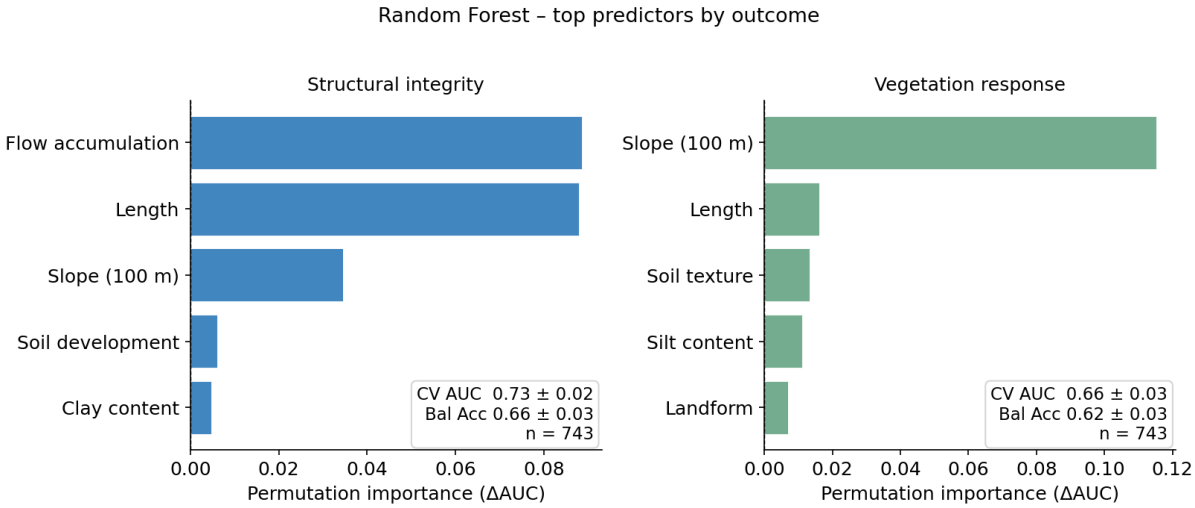
This document summarises each registered figure for Paper 1: the figure image is shown above a two-column table giving the filename, last-updated date, the statistical test applied, and a brief interpretation of the key result.

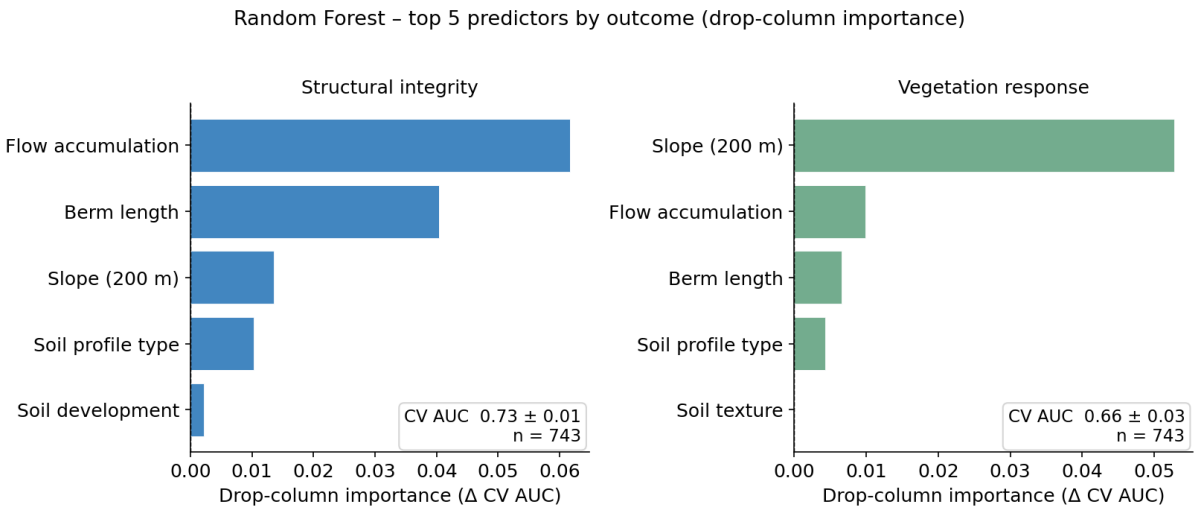
Figure 1



File: `fig1_rf_permutation_importance.png` *Updated:* 2026-02-28 00:00

Field	Content
Figure	Figure 1
File	`fig1_rf_permutation_importance.png`
Updated	2026-02-28 00:00
Statistical test	Random Forest classifier with 5-fold cross-validation; permutation importance computed by randomly shuffling each predictor and measuring the decrease in AUC. Top 5 predictors shown for each outcome.
Interpretation	The top predictors for structural integrity are flow accumulation, berm length, slope (100 m and 200 m), and soil texture. For vegetation response, the top predictors are landform, soil texture, flow accumulation, soil development, and berm length. Both outcomes are strongly associated with berm length and landscape position (flow accumulation), while vegetation response is more influenced by soil and landform characteristics than structural integrity.

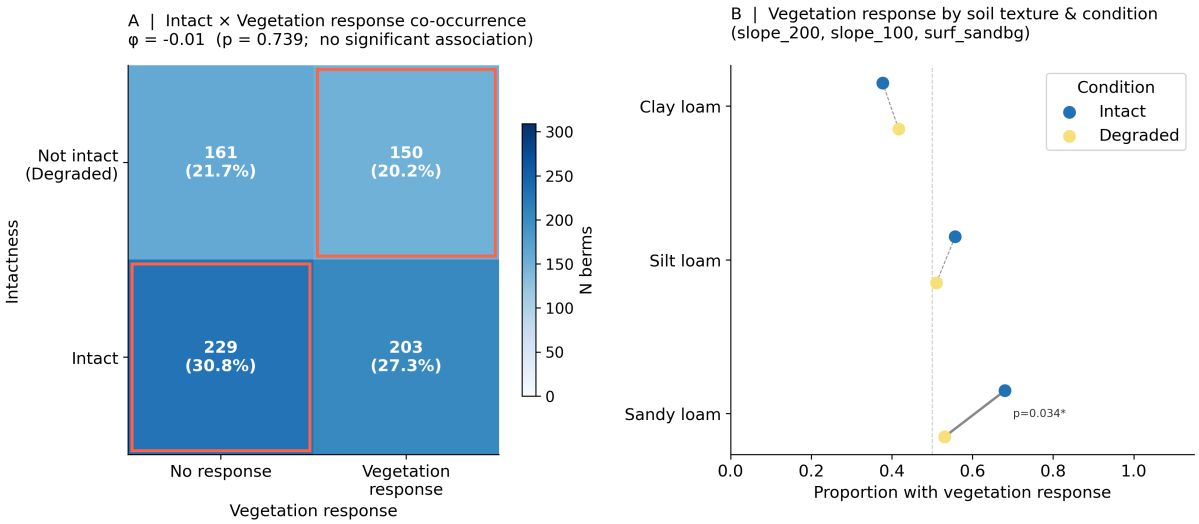
Figure 1 (alternate)



File: `fig1_alt.png` *Updated:* 2026-02-28 00:00

Field	Content
Figure	Figure 1 (alternate)
File	`fig1_alt.png`
Updated	2026-02-28 00:00
Statistical test	Random Forest classifier with 5-fold cross-validation; drop-column importance computed by comparing baseline CV AUC (with all predictors) to CV AUC with each predictor removed. Top 5 predictors shown for each outcome.
Interpretation	The top predictors for structural integrity are berm length, slope (200 m), slope (100 m), flow accumulation, and soil texture. For vegetation response, the top predictors are landform, soil texture, flow accumulation, berm length, and soil development. Both outcomes show strong associations with berm length and landscape position (flow accumulation), while vegetation response is more strongly influenced by soil and landform characteristics.

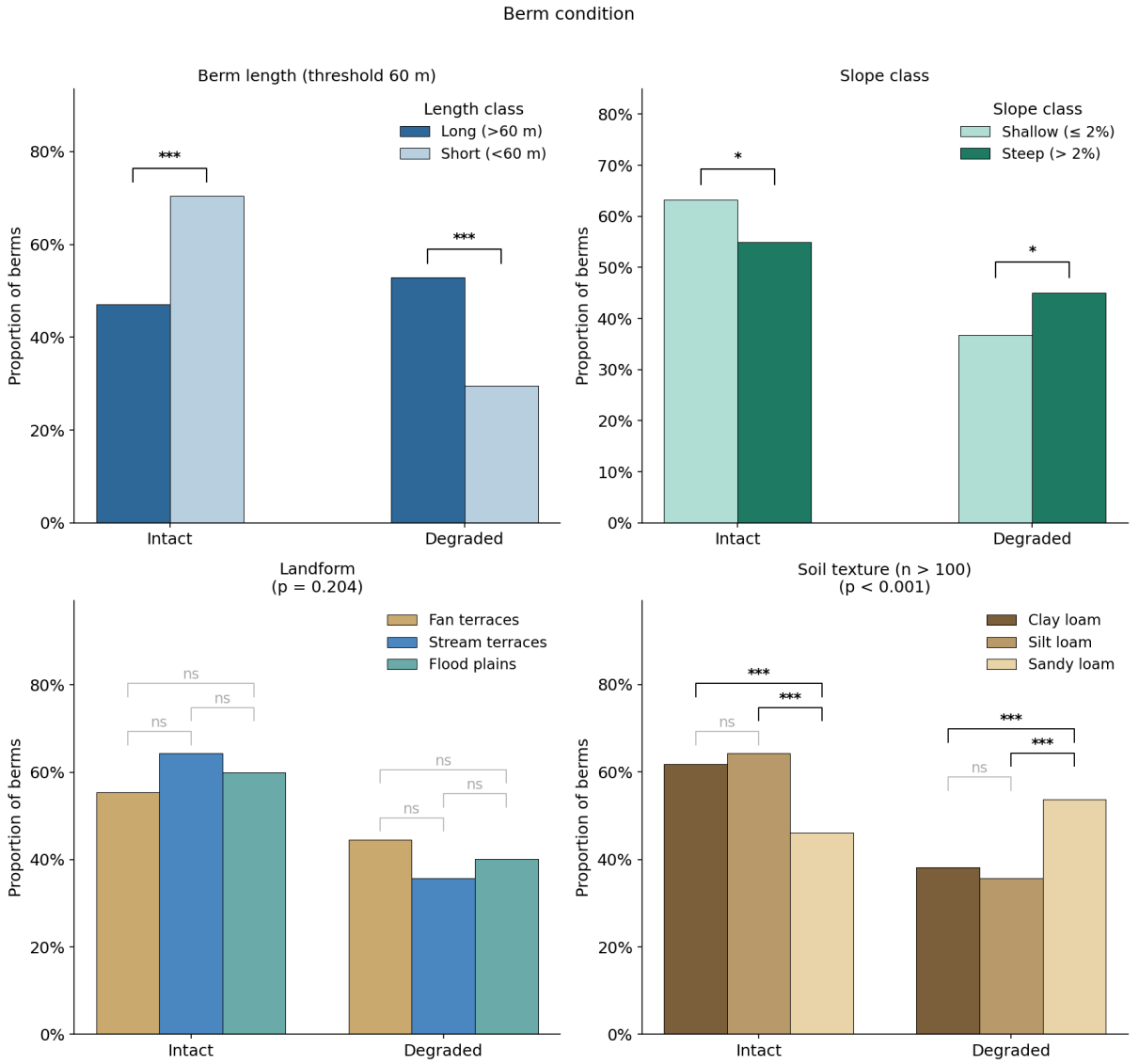
Figure 2



File: `fig2_geomorph_soil_intact_effective.png` *Updated:* 2026-02-27 08:45

Field	Content
Figure	Figure 2
File	`fig2_geomorph_soil_intact_effective.png`
Updated	2026-02-27 08:45
Statistical test	Panel A: Pearson χ^2 (φ coefficient) for a 2×2 Intact × Vegetation-response contingency table ($N = 743$). Panel B: two-sided Fisher's exact test comparing proportions between Intact and Degraded berms within each soil texture class.
Interpretation	Berm intactness and vegetation response are statistically independent ($\varphi = -0.01$, $p = 0.739$), indicating that intact and degraded berms produce vegetation responses at similar overall rates. The exception is Sandy loam, where intact berms show significantly higher vegetation response than degraded berms ($p = 0.034$).

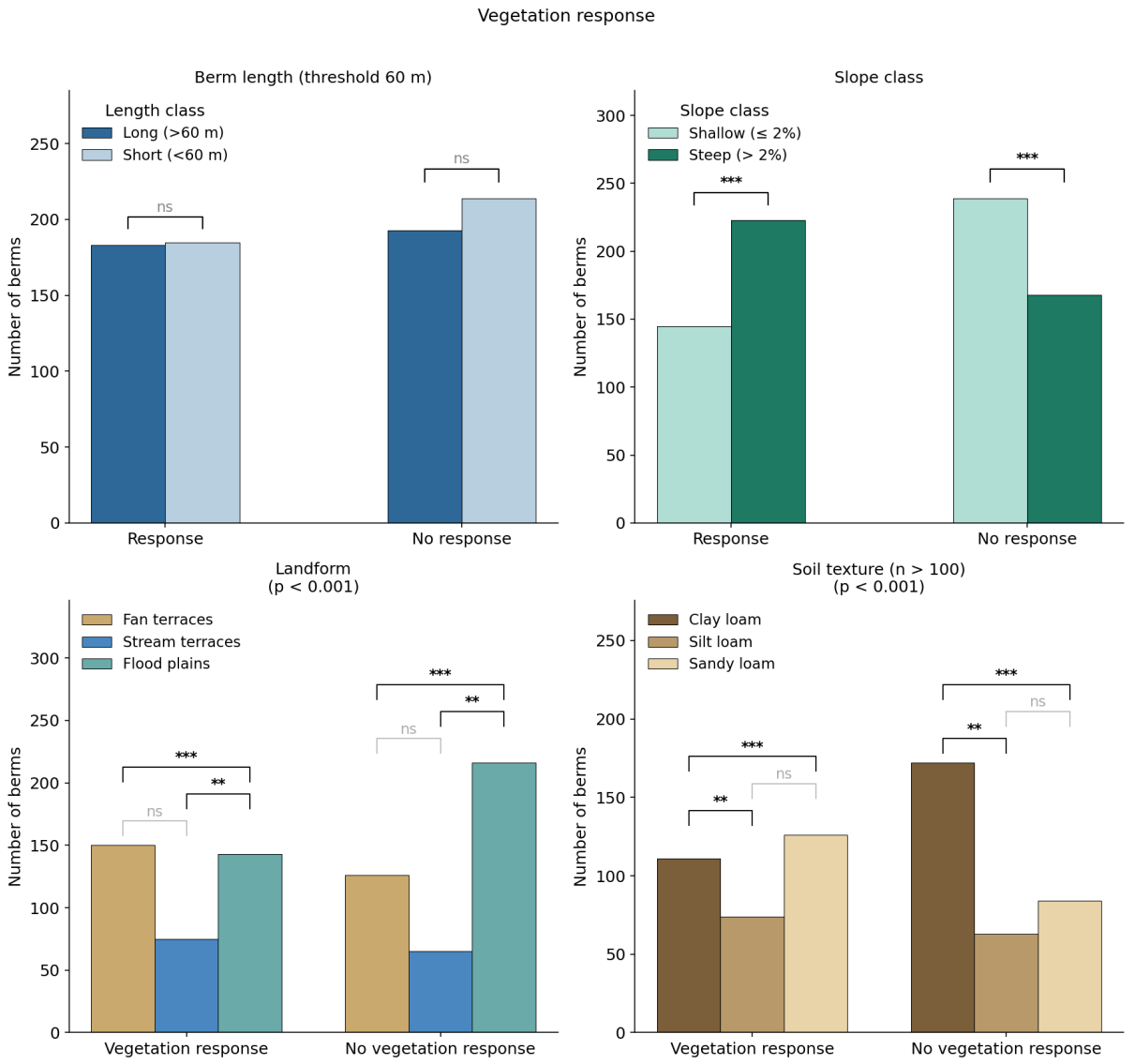
Figure 3



File: `fig3_berm_condition.png` *Updated:* 2026-02-27 09:39

Field	Content
Figure	Figure 3
File	`fig3_berm_condition.png`
Updated	2026-02-27 09:39
Statistical test	Two-category panels (length, slope) use a one-sided Fisher's exact test; multi-category panels (landform, texture) use pairwise Fisher's exact tests with the overall χ^2 p-value shown in the panel title.
Interpretation	Berm condition is strongly associated with berm length — shorter berms are markedly more likely to be intact ($p < 0.001$) — and modestly associated with slope class ($p < 0.05$). Landform does not significantly predict condition ($p = 0.204$), but soil texture does ($p < 0.001$).

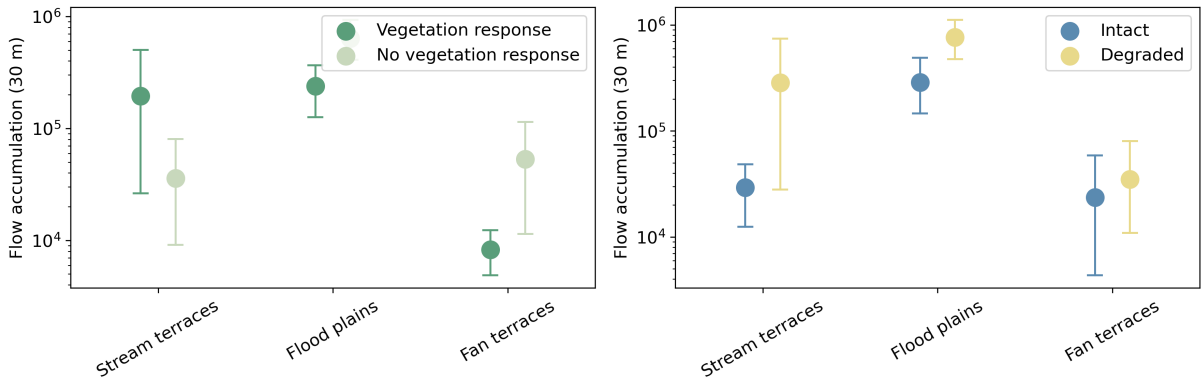
Figure 4



File: `fig4_vegetation_response.png` *Updated:* 2026-02-27 08:54

Field	Content
Figure	Figure 4
File	`fig4_vegetation_response.png`
Updated	2026-02-27 08:54
Statistical test	Two-category panels (length, slope) use a one-sided Fisher's exact test; multi-category panels (landform, texture) use pairwise Fisher's exact tests with the overall χ^2 p-value shown in the panel title.
Interpretation	Vegetation response is significantly associated with slope class ($p < 0.001$), landform ($p < 0.001$), and soil texture ($p < 0.001$), but not with berm length alone. Steep-slope berms and flood-plain berms show the highest rates of vegetation response.

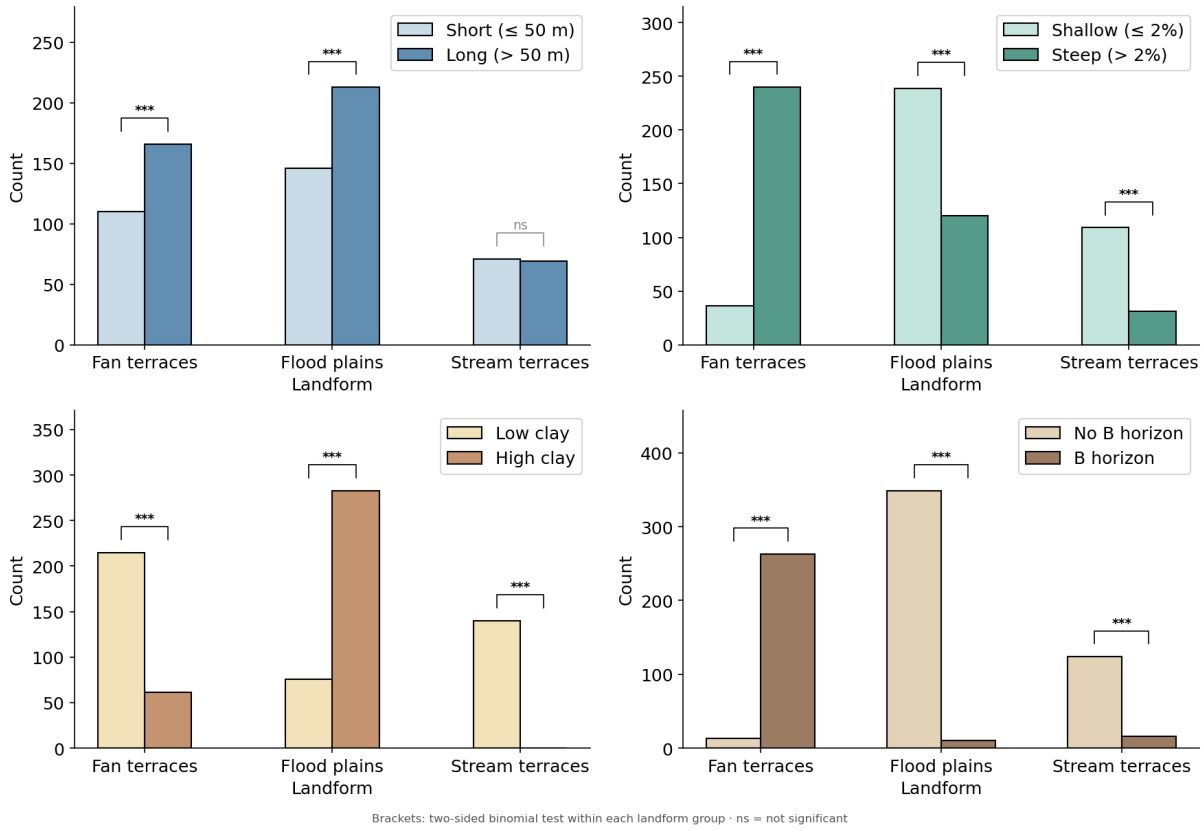
Supplementary Figure 1



File: `SI_fig1_fa30_pointplot_by_landform.png` *Updated:* 2026-02-28 20:12

Field	Content
Figure	Supplementary Figure 1
File	`SI_fig1_fa30_pointplot_by_landform.png`
Updated	2026-02-28 20:12
Statistical test	Point plots show mean ± 95% CI of 30 m flow accumulation (log scale) by landform, grouped by vegetation response (Panel A) and structural condition (Panel B).
Interpretation	Flood-plain berms experience the highest contributing flow accumulation regardless of outcome, while fan-terrace berms receive the least. Within each landform, effective berms tend to have similar or higher flow accumulation than ineffective berms, and degraded berms tend toward higher values than intact berms, consistent with greater hydraulic stress on degraded structures.

Supplementary Figure 2



File: `SI_landform_by_predictors.png` *Updated:* 2026-02-27 08:43

Field	Content
Figure	Supplementary Figure 2
File	`SI_landform_by_predictors.png`
Updated	2026-02-27 08:43
Statistical test	Two-sided binomial test (H_0: 50/50 split) applied within each landform group for each predictor category (length, slope, clay content, soil development); labels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns $p \geq 0.05$.
Interpretation	Predictors are non-randomly distributed across landforms — high-clay soils concentrate on flood plains, steep slopes on fan terraces, and B-horizon development on fan and stream terraces — confirming that landform is a meaningful stratification variable and motivating its inclusion as a covariate.

Supplementary Tables

Supplementary Table 1: Univariate GLM Rankings for Vegetation Response Predictors

These tables present univariate predictor rankings using binomial generalized linear models (GLMs) to identify the most informative individual predictors for each outcome. For each predictor, we report:

- **McFadden pseudo-R^2**: measures improvement in log-likelihood relative to the null model (higher values indicate better fit)
- **Tjur R^2**: measures separation in predicted probabilities between outcome classes
- **CV AUC**: 5-fold cross-validated area under the ROC curve (higher values indicate better predictive performance)
- **LRT p-value**: likelihood ratio test comparing the model to the null model

These univariate rankings complement the multivariate Random Forest analysis (Figure 1) by showing how each predictor performs in isolation, helping to identify the strongest individual associations before accounting for interactions and non-linear effects.

Top 15 predictors for Vegetation Response

Predictor	Type	n	McFadden R²	Tjur R²	CV AUC	
$slope_100$	numeric	743	0.013867154781099345	0.019498550165042927	0.6107068049321571	0.0
ParentMaterial	categorical	621	0.025057457836702324	0.034236635191075226	0.584937752081707	8.
TypicalProfile	categorical	743	0.048637827294604086	0.06230113238212309	0.5767014909972656	4.
$claytotal_r$	numeric	743	0.011556320876656034	0.01593142029487865	0.5744804725790642	0.
Texture	categorical	743	0.02686778167367987	0.0368558311104078	0.5731014290873446	0
Landform	categorical	743	0.014500971048470146	0.019978554830673956	0.5634133003147086	0.
$Soil_Development$	categorical	743	0.009346868549408804	0.012924295622223025	0.5550152195222617	0.
FA_30_max	numeric	743	0.007735041724707625	0.01010547341436202	0.5334117525666822	0
$sandtotal_r$	numeric	743	0.00223048645024837	0.0030980378696865984	0.5107916731156168	(
$silttotal_r$	numeric	743	3.294963721300892e-05	4.582842573047641e-05	0.47741345508951144	
$Shape_Leng$	numeric	743	0.0002677206852584879	0.00036744652696196756	0.47635814889336014	

Supplementary Table 2: Univariate GLM Rankings for Structural Integrity Predictors

Top 15 predictors for Structural Integrity

Predictor	Type	n	McFadden R²	Tjur R²	CV AUC	
Shape$_{L}eng$	numeric	743	0.03767771093029071	0.05107200775672993	0.6609540726725196	6.85
FA$_{3}0_{m}ax$	numeric	743	0.007872094338936542	0.011615310288193625	0.6604547585993379	0.0
TypicalProfile	categorical	743	0.07173339502474918	0.09065009666025647	0.6294997805280549	8.07
Texture	categorical	743	0.03149295029471921	0.04083804827769344	0.5956934943916603	4.39
claytotal$_{r}$	numeric	743	0.012698208574921432	0.017405849199885748	0.578667953113073	0.00
sandtotal$_{r}$	numeric	743	0.012359430475675781	0.016881463008425746	0.5624065909745088	0.00
ParentMaterial	categorical	621	0.029111490147361807	0.039961529370305726	0.5481821946169771	1.71
silttotal$_{r}$	numeric	743	0.007524600169383944	0.010208759809466561	0.5471468852434784	0.00
Soil$_{D}evelopment$	categorical	743	0.004610474274106102	0.006288408331316053	0.5387198811197053	0.0
Landform	categorical	743	0.003626903547485183	0.004905119676970271	0.5376871414076836	0.1
slope$_{1}$00	numeric	743	1.063922026789399e-05	1.3955032340029483e-05	0.4840634419688666	0.9